

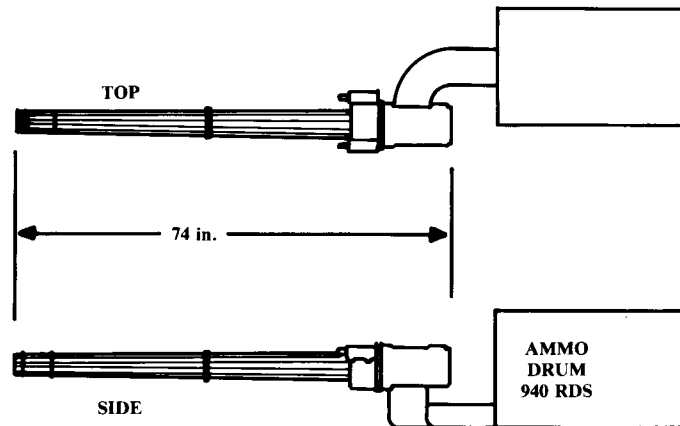


Fig. 9.9 M61 "VULCAN" gun.

An air-to-air gun such as the M61A1 can produce a recoil force on the order of two tons. A large anti-tank gun such as the GAU-8 used in the A-10 can produce a recoil force five times greater. To avoid a sudden yawing motion from firing, guns should be located as near as possible to the centerline of the aircraft. On the A-10, the nose landing gear is offset to one side to allow the gun to be exactly on the centerline. This extreme is not necessary for the smaller air-to-air guns.

When a gun is fired, it produces a bright flash and a large cloud of smoke. The gun muzzle should be located so that these do not obscure the pilot's vision. Also, being very noisy, a gun should be located away from the cockpit.

The cloud of smoke produced by a gun can easily stall a jet engine if sucked into the inlet. This should also be considered when locating a gun.

10

PROPULSION AND FUEL SYSTEM INTEGRATION

10.1 INTRODUCTION

This section treats the integration and layout of the propulsion system into the overall vehicle design, not the calculation of installed propulsion performance. Propulsion analysis methods are covered in Chapter 13.

To develop the propulsion system layout it is necessary to know the actual dimensions and installation requirements of the engine as well as its supporting equipment such as inlet ducts, nozzles, or propellers. Also, the fuel system including the fuel tanks must be defined.

10.2 PROPULSION SELECTION

Figure 10.1 illustrates the major options for aircraft propulsion. All aircraft engines operate by compressing outside air, mixing it with fuel, burning the mixture, and extracting energy from the resulting high-pressure hot gases. In a piston-prop, these steps are done intermittently in the cylinders via the reciprocating pistons. In a turbine engine, these steps are done continuously, but in three distinct parts of the engine.

The piston-prop was the first form of aircraft propulsion. By the dawn of the jet era, a 5500-hp piston-prop engine was in development. Today piston-props are mainly limited to light airplanes and some agricultural aircraft.

Piston-prop engines have two advantages. They are cheap, and they have the lowest fuel consumption. However, they are heavy and produce a lot of noise and vibration. Also, the propeller loses efficiency as the velocity increases.

The turbine engine consists of a "compressor," a "burner," and a "turbine." These separately perform the three functions of the reciprocating piston in a piston engine.

The compressor takes the air delivered by the inlet system and compresses it to many times atmospheric pressure. This compressed air passes to the burner, where fuel is injected and mixed with the air and the resulting mixture ignited.

The hot gases could be immediately expelled out the rear to provide thrust, but are first passed through a turbine to extract enough mechanical power to drive the compressor. It is interesting to note that one early jet engine used a separate piston engine to drive the compressor.

There are two types of compressors. The centrifugal compressor relies upon centrifugal force to "fling" the air into an increasingly narrow chan-

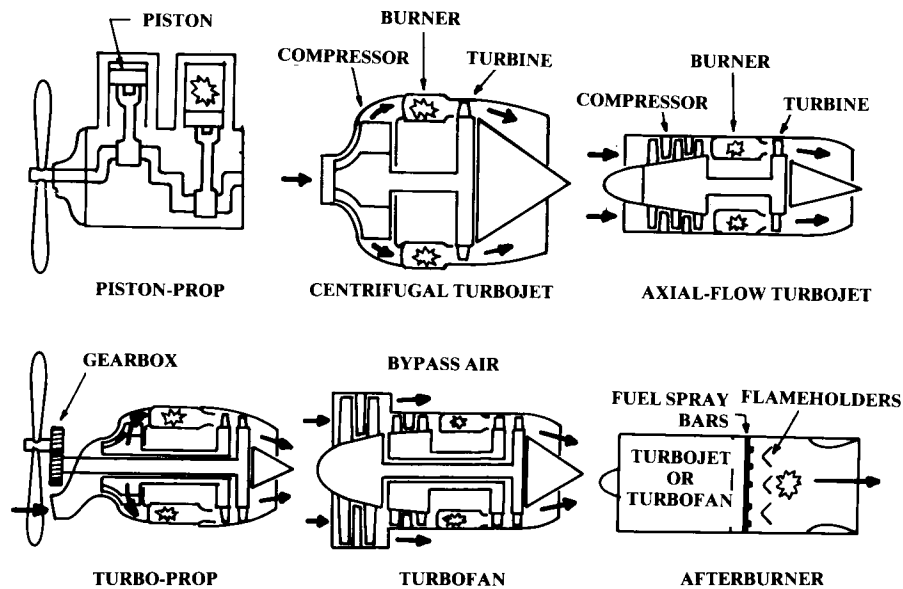


Fig. 10.1 Propulsion system options.

nel, which raises the pressure. In contrast, an axial compressor relies upon blade aerodynamics to force the air into an increasingly narrow channel. An axial compressor typically has about six to ten “stages,” each of which consists of a “rotor” (i.e., rotating) disk of blades and a “stator” (i.e., stationary) disk of blades. The rotors tend to swirl the air, so the stators are used to remove the swirl.

The axial compressor, relying upon blade aerodynamics, is intolerant to distortions in the incoming air such as swirl or pressure variations. These distortions can stall the blades, causing a loss of compression and a possible engine flame-out.

The centrifugal compressor is much more forgiving of inlet distortion, but causes the engine to have a substantially higher frontal area, which increases aircraft drag. Also, a centrifugal compressor cannot provide as great a pressure increase (pressure ratio) as an axial compressor. Several smaller turbine engines use a centrifugal compressor behind an axial compressor to attempt to get the best of both types.

The turboprop and turbofan engines both use a turbine to extract mechanical power from the exhaust gases. This mechanical power is used to accelerate a larger mass of outside air, which increases efficiency at lower speeds.

For the turboprop engine, the outside air is accelerated by a conventional propeller. The “prop-fan” or “unducted fan” is essentially a turboprop with an advanced aerodynamics propeller capable of near-sonic speeds.

For the turbofan engine, the air is accelerated with a ducted fan of one or several stages. This accelerated air is then split, with part remaining in the

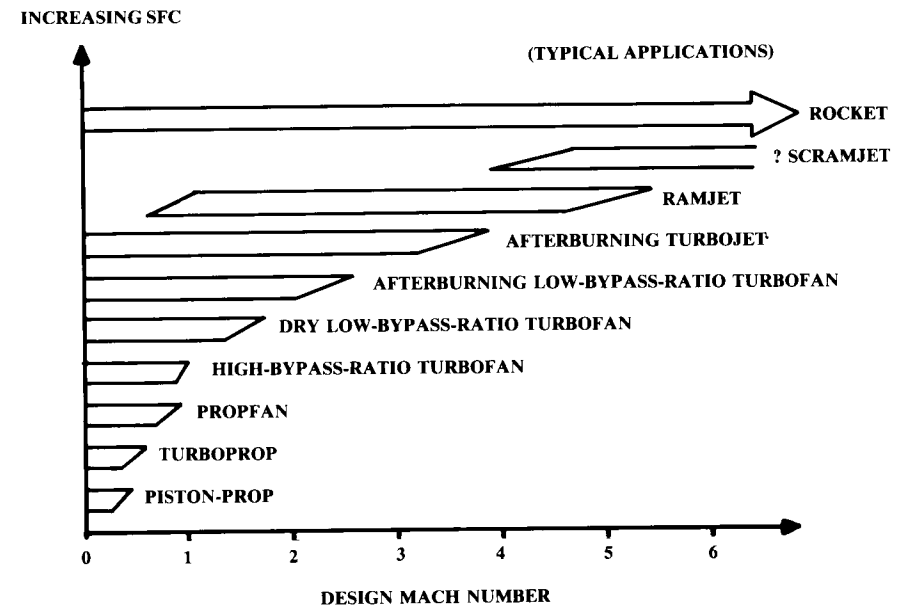


Fig. 10.2 Propulsion system speed limits.

engine for further compression and burning, and the remainder being “bypassed” around the engine to exit unburned.

The “bypass ratio” is the mass-flow ratio of the bypassed air to the air that goes into the core of the engine. Bypass ratio ranges from as high as 6 to as low as 0.25 (the so-called “leaky turbojet”).

The ideal turbine engine would inject enough fuel to completely combust all of the compressed air, producing maximum thrust for a given engine size. Unfortunately, this “stoichiometric” air/fuel mixture ratio of about 15 to 1 produces temperatures far greater than the capabilities of known materials, and would therefore burn up the turbine blades.

To lower the temperature seen by the turbine blades, excess air is used. Currently engines are limited to a turbine temperature of about 2000–2500°F, which requires an air/fuel mixture ratio of about 60 to 1. Thus, only about a quarter of the captured and compressed air is actually used for combustion. The exhaust is 75% unused hot air.

If fuel is injected into this largely-uncombusted hot air, it will mix and burn. This will raise the thrust as much as a factor of two, and is known as “afterburning.” Unfortunately, afterburning is inefficient in terms of fuel usage. The fuel flow required to produce a pound of thrust in afterburner is approximately double that used to produce a pound of thrust during normal engine operations.

Due to the high temperatures produced, afterburning must be done downstream of the turbine. Also, it is usually necessary to divert part of the compressor air to cool the walls of the afterburner and nozzles. Addition of

an afterburner will approximately double the length of a turbojet or turbofan engine.

If the aircraft is traveling fast enough, the inlet duct alone will compress the air enough to burn if fuel is added. This is the principle of a "ramjet." Ramjets must be traveling at above Mach 3 to become competitive with a turbojet in terms of efficiency.

A "scramjet" is a ramjet that can operate with supersonic internal flow and combustion. Scramjets are largely unproven as of this writing, and are probably suitable only for operation above Mach 5 or 6. Ramjets and scramjets require some other form of propulsion for takeoff and acceleration to the high Mach numbers they require for operation.

The selection of the type of propulsion system—piston-prop, turboprop, turbofan, turbojet, ramjet—will usually be obvious from the design requirements. Aircraft maximum speed limits the choices, as shown in Fig. 10.2. In most cases there is no reason to select a propulsion system other than the lowest on the chart for the design Mach number. Fuel-consumption trends have been shown in Fig. 3.3.

The choice between a piston-prop and a turboprop can depend upon several additional factors. The turboprop uses more fuel than a piston prop of the same horsepower, but is substantially lighter and more reliable. Also, turboprops are usually quieter. For these reasons turbine engines have largely replaced piston engines for most helicopters, business twins, and short-range commuter airplanes regardless of design speed. However, piston-props are substantially cheaper and will likely remain the only choice for light aircraft for a long time.

10.3 JET-ENGINE INTEGRATION

Engine Dimensions

If the aircraft is designed using an existing, off-the-shelf engine, the dimensions are obtained from the manufacturer. If a "rubber" engine is being used, the dimensions for the engine must be obtained by scaling from some nominal engine size by whatever scale factor is required to provide the desired thrust. The nominal engine can be obtained by several methods.

In the major aircraft companies, designers can obtain estimated data for hypothetical "rubber" engines from the engine companies. This data is presented for a nominal engine size, and precise scaling laws are provided. Appendix A.4 provides data for several hypothetical advanced engines.

Better yet, engine companies sometimes provide a "parametric deck," a computer program that will provide performance and dimensional data for an arbitrary advanced-technology engine based upon inputs such as bypass ratio, overall pressure ratio, and turbine-inlet temperature. This kind of program, which provides great flexibility for early trade studies, goes beyond the scope of this book.

Another method for defining a nominal engine assumes that the new engine will be a scaled version of an existing one, perhaps with some performance improvement due to the use of newer technologies. For example, in designing a new fighter one could start with the dimensions and performance charts of the P&W F-100 turbofan, which powers the F-15 and F-16.

To approximate the improvements due to advanced technologies, one could assume, say, a 10 or 20% reduction in fuel consumption and a similar reduction in weight. This would reflect the better materials, higher operating temperatures, and more efficient compressors and turbines that could be built today.

Figure 10.3 illustrates the dimensions that must be scaled from the nominal engine. The scale factor "SF" is the ratio between the required thrust and the actual thrust of the nominal engine. Equations (10.1–10.3) show how length, diameter, and weight vary with the scale factor for the typical jet engine.

$$L = L_{\text{actual}}(\text{SF})^{0.4} \quad (10.1)$$

$$D = D_{\text{actual}}(\text{SF})^{0.5} \quad (10.2)$$

$$W = W_{\text{actual}}(\text{SF})^{1.1} \quad (10.3)$$

Although statistically derived, these equations make intuitive sense. Thrust is roughly proportional to the mass flow of air used by the engine, which is related to the cross-sectional area of the engine. Since area is proportional to the square of the diameter, it follows that the diameter should be proportional to the square root of the thrust scale-factor.

Note the engine-accessories package beneath the engine. The accessories include fuel pumps, oil pumps, power-takeoff gearboxes, and engine control boxes. The location and size of the accessory package varies widely for different types of engines. In the absence of a drawing, the accessory package can be assumed to extend below the engine to a radius of about 20–40% greater than the engine radius. On some engines these accessories have been located in the compressor spinner or other places.

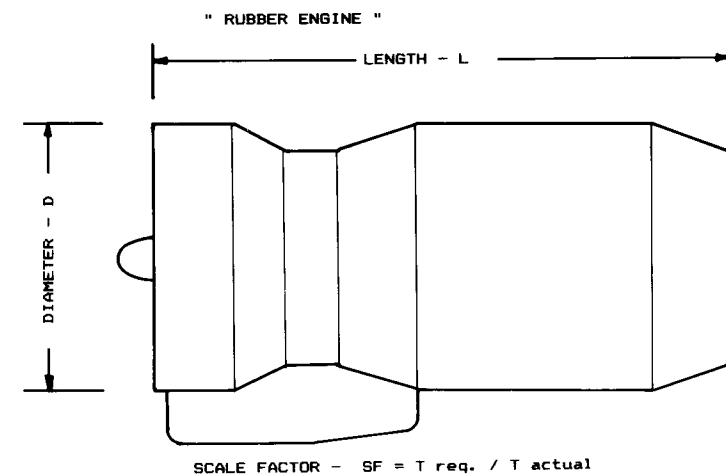


Fig. 10.3 Engine scaling.

If a parametric deck is unavailable, and no existing engines come close enough to the desired characteristics to be rubberized and updated as described above, then a parametric statistical approach can be used to define the nominal engine.

Equations (10.4–10.15) define two first-order statistical jet-engine models based upon data from Ref. 1. One model is for subsonic nonafterburning engines such as found on commercial transports, and covers a bypass-ratio range from zero to about 6. The other model is for afterburning engines for supersonic fighters and bombers ($M < 2.5$), and includes bypass ratios from zero to just under 1.

Nonafterburning engines:

$$W = 0.084T^{1.1}e^{(-0.045 \text{ BPR})} \quad (10.4)$$

$$L = 2.22T^{0.4}M^{0.2} \quad (10.5)$$

$$D = 0.393T^{0.5}e^{(0.04 \text{ BPR})} \quad (10.6)$$

$$\text{SFC}_{\max T} = 0.67e^{(-0.12 \text{ BPR})} \quad (10.7)$$

$$T_{\text{cruise}} = 0.60T^{0.9}e^{(0.02 \text{ BPR})} \quad (10.8)$$

$$\text{SFC}_{\text{cruise}} = 0.88e^{(-0.05 \text{ BPR})} \quad (10.9)$$

Afterburning engines:

$$W = 0.063T^{1.1}M^{0.25}e^{(-0.81 \text{ BPR})} \quad (10.10)$$

$$L = 3.06T^{0.4}M^{0.2} \quad (10.11)$$

$$D = 0.288T^{0.5}e^{(0.04 \text{ BPR})} \quad (10.12)$$

$$\text{SFC}_{\max T} = 2.1e^{(-0.12 \text{ BPR})} \quad (10.13)$$

$$T_{\text{cruise}} = 1.6T^{0.74}e^{(0.023 \text{ BPR})} \quad (10.14)$$

$$\text{SFC}_{\text{cruise}} = 1.04e^{(-0.186 \text{ BPR})} \quad (10.15)$$

where

W = weight
 T = takeoff thrust
 BPR = bypass ratio
 M = max Mach number
 Cruise is at 36,000 ft and $0.9M$.

These equations represent a very unsophisticated model for initial estimation of engine dimensions. They should not be applied beyond the given bypass-ratio and speed ranges. Also, these equations represent today's state of the art. Improvement factors should be applied to approximate future

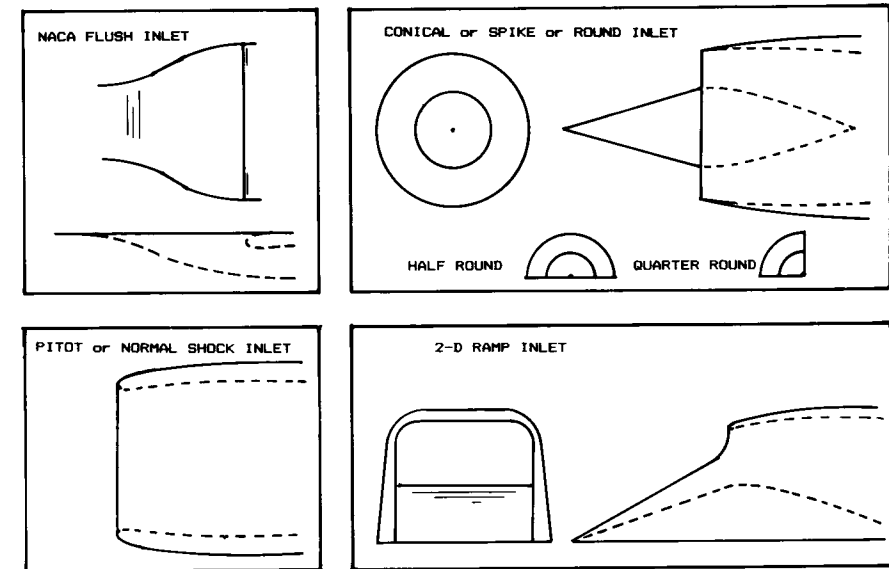


Fig. 10.4 Inlet types.

engines. For a post-1995 engine this author recommends, as a crude approximation, a 20% reduction in SFC, weight, and length for a given maximum thrust.

Reference 46 is recommended for the theory and practice of jet-engine design.

Inlet Geometry

Turbojet and turbofan engines are incapable of efficient operation unless the air entering them is slowed to a speed of about Mach 0.4–0.5. This is to keep the tip speed of the compressor blades below sonic speed relative to the incoming air. Slowing down the incoming air is the primary purpose of an inlet system.

The installed performance of a jet engine greatly depends upon the air-inlet system. The type and geometry of the inlet and inlet duct will determine the pressure loss and distortion of the air supplied to the engine, which will affect the installed thrust and fuel consumption. Roughly speaking, a 1% reduction in inlet pressure recovery (total pressure delivered to the engine divided by freestream total pressure) will reduce thrust by about 1.3%.

Also, the inlet's external geometry including the cowl and boundary-layer diverter will greatly influence the aircraft drag. As discussed in Chapter 13, this drag due to propulsion is counted as a reduction in the installed thrust.

There are four basic types of inlet, as shown in Fig. 10.4. The NACA flush inlet was used by several early jet aircraft but is rarely seen today for aircraft propulsion systems due to its poor pressure recovery (i.e., large losses). At the subsonic speeds for which the NACA inlet is suitable, a

pitot-type inlet will have virtually 100% pressure recovery vs about 90% for a well-designed NACA inlet. However, the NACA inlet tends to reduce aircraft wetted area and weight if the engine is in the fuselage.

The NACA inlet is regularly used for applications in which pressure recovery is less important, such as the intakes for cooling air or for turbine-powered auxiliary power units. The BD-5J, a jet version of the BD-5 homebuilt, used the NACA inlet, probably to minimize the redesign effort.

Figure 10.5 and Table 10.1 provide dimensions for laying out a good NACA flush inlet. This inlet will provide as high as 92% pressure recovery when operating at a mass flow ratio of 0.5 (i.e., air mass flow through inlet is 0.5 times the mass flow through the same cross-sectional area in the freestream).

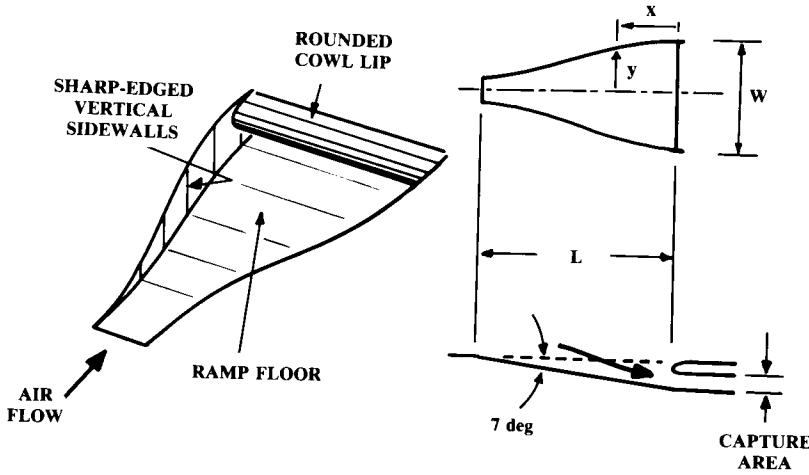


Fig. 10.5 Flush inlet geometry.

Table 10.1 Flush inlet wall geometry

x/L	$\frac{y}{W/2}$
1.0	0.083
0.9	0.160
0.8	0.236
0.7	0.313
0.6	0.389
0.5	0.466
0.4	0.614
0.3	0.766
0.2	0.916
0.1	0.996
0.0	1.000

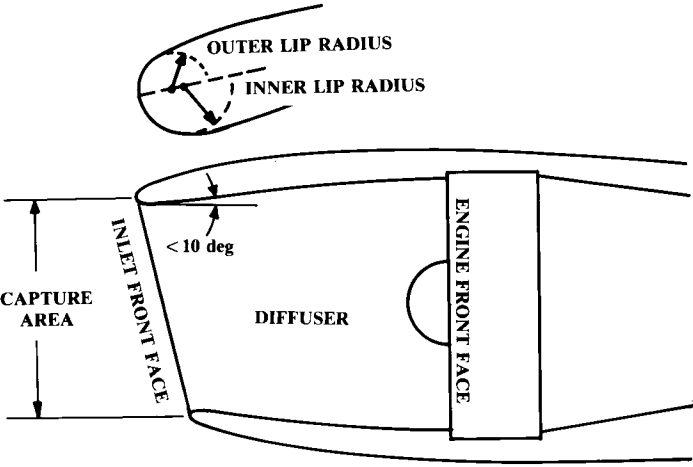


Fig. 10.6 Pitot (normal shock) inlet layout.

The pitot inlet is simply a forward-facing hole. It works very well subsonically and fairly well at low supersonic speeds. It is also called a “normal shock inlet” when used for supersonic flight (“normal” meaning perpendicular in this case). Figure 10.6 gives design guidance for pitot inlets.

The cowl lip radius has a major influence upon engine performance and aircraft drag. A large lip radius tends to minimize distortion, especially at high angles of attack and sideslip. Also, a large lip radius will readily accommodate the additional air required for takeoff thrust, when the ram air effect is small. However, a large lip radius will produce shock-separated flow on the outside of the inlet as the speed of sound is approached, and that greatly increases the drag.

For supersonic jets, the cowl lip should be nearly sharp. Typically the lip radius will be about 3–5% of the inlet front face radius. For subsonic jets, the lip radius ranges from 6–10% of the inlet radius.

To minimize distortion the lip radius on a subsonic inlet is frequently greater on the inside than the outside, with perhaps an 8% inner radius and a 4% outer radius. Also, a number of aircraft have a lip radius on the lower part of the inlet up to 50% greater than that on the upper lip. This reduces the effects of angle of attack during takeoff and landing.

Note that the inlet front face may not be perpendicular to the engine axis. The desired front-face orientation depends upon the location of the inlet and the aircraft’s angle-of-attack range. Normally the inlet should be about perpendicular to the local flow direction during cruise. If the aircraft is to operate at large angles of attack, it may be desirable to compromise between these angles and the angle at cruise.

The remaining inlet types shown in Fig. 10.4 are for supersonic aircraft, and offer improvements over the performance of the normal shock inlet at higher supersonic speeds. The conical inlet (also called a spike, round, or axisymmetric inlet) exploits the shock patterns created by supersonic flow

over a cone. Similarly, the two-dimensional ramp inlet (also called a “*D*-inlet”) uses the flow over a wedge.

The spike inlet is typically lighter and has slightly better pressure recovery (1.5%), but has higher cowl drag and involves much more complicated mechanisms to produce variable geometry. The ramp inlet tends to be used more for speeds up to about Mach 2, while the spike inlet tends to be used above that speed.

Any inlet must slow the air to about half the speed of sound before it reaches the engine. The final transition from supersonic to subsonic speed always occurs through a normal shock. The pressure recovery through a shock depends upon the strength of the shock, which is related to the speed reduction through the shock.

In other words, a normal shock used to slow air from Mach 2 down to subsonic speeds will have a far worse pressure recovery than a normal shock used to slow the air from Mach 1.1 to subsonic speeds (72% vs 99.9%). For this reason a normal-shock inlet is rarely used for prolonged operation above Mach 1.4.

An oblique shock, however, does not reduce the air speed all the way to subsonic. The speed reduction and pressure recovery through an oblique shock depends upon the angle of the wedge or cone used to establish the shock. For example, a 10-deg wedge in Mach 2 flow creates an oblique shock at 39 deg that reduces the flow speed to Mach 1.66 (see Appendix A.2). This gives a pressure loss of only 1.4% (i.e., pressure recovery of 98.6%).

If the Mach 1.66 air downstream of this oblique shock is then run into a normal shock inlet, it will slow to Mach 0.65, with a pressure recovery of 87.2%. The total pressure recovery from Mach 2 to subsonic speed is 98.6 times 87.2, or 86%. Thus, use of an oblique shock before the normal shock has improved pressure recovery for this example Mach 2 inlet from 72% to 86%. (Note that this is far from optimal. A well-designed Mach 2 inlet with one oblique shock will approach a 95% pressure recovery.)

This illustrates the principle of the external-compression inlet shown in Fig. 10.7. The above example is a two-shock system, one external and one normal. The greater the number of oblique shocks employed, the better the pressure recovery.

The theoretical optimal is the isentropic ramp inlet, which corresponds to infinitely-many oblique shocks and produces a pressure recovery of 100% (ignoring friction losses). The pure isentropic ramp inlet works properly at only its design Mach number, and is seen only rarely. However, isentropic ramps are frequently used in combination with flat wedge ramps, such as on the Concorde SST.

Figure 10.8 illustrates a typical three-shock external-compression inlet. This illustration could be a side view of a 2-D inlet or a section view through a spike inlet. Note that the second ramp has a variable angle, and can collapse to open a larger duct opening for subsonic flight.

Some form of boundary-layer bleed is required on the ramp to prevent shock-induced separation on the ramp. The bled air is usually dumped overboard out a rearward-facing hole above the inlet duct.

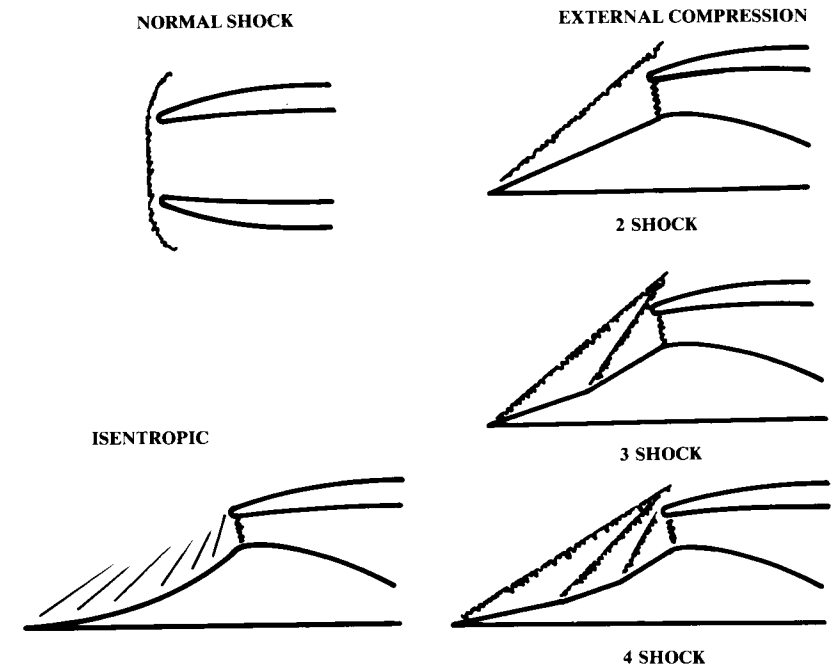


Fig. 10.7 Supersonic inlets—external shocks.

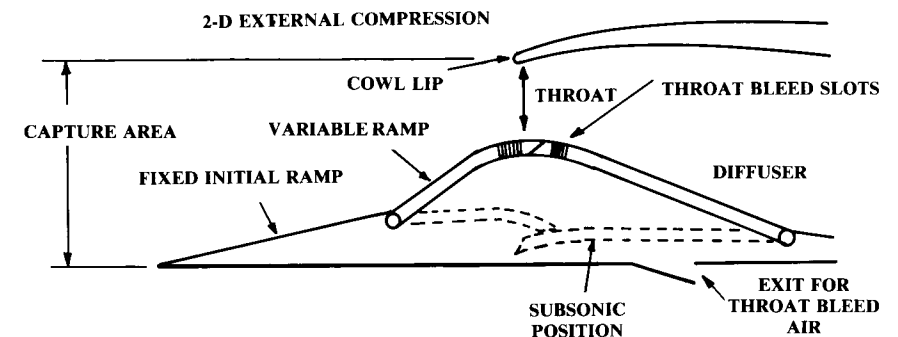


Fig. 10.8 Variable inlet geometry.

Not shown are “suck-in” or “bypass” doors in the diffuser section that may be required to provide extra air to the engine for takeoff or get rid of excess air during high-speed operation.

For initial layout, the overall length of the external portion of the inlet can be estimated by assuming an initial ramp angle (10–20 deg) and determining the shock angle for the design Mach number using standard shock charts such as in Appendix A.2. The cowl lip should be placed just aft of the shock.

The throat area should be about 70–80% of the engine front-face area.

The speed limitation on external compression inlets is due to the flow turning angle introduced by the shocks. A wedge turns the flow parallel to the wedge angle, while a cone turns the flow to an angle slightly less than the cone angle.

At speeds approaching Mach 3, the required oblique shocks to obtain good pressure recovery will introduce a total flow turning of about 40 deg. This air must be turned by the outside cowl lip back to the freestream direction, which may not be possible without either separation or an excessively large lip radius that will increase aircraft drag.

One form of inlet system introduces no outside flow turning: the internal compression inlet, as shown in Fig. 10.9. In this inlet a pair of inward-facing ramps produce oblique shocks that cross upstream of the final normal shock.

This form of shock system can be very efficient when operating properly at its design Mach number. However, this inlet must be “started.” If it is simply placed into supersonic flow, a normal shock will form across its front. To start the inlet and produce the efficient shock structure shown in Fig. 10.9, it is necessary to “suck” the normal shock down to the throat by opening doors downstream. Once formed, the desired shock structure is unstable. Any deviation in flow condition, such as temperature, pressure, or angle of attack can cause an “unstart” in which the normal shock pops out of the duct. This can stall the engine.

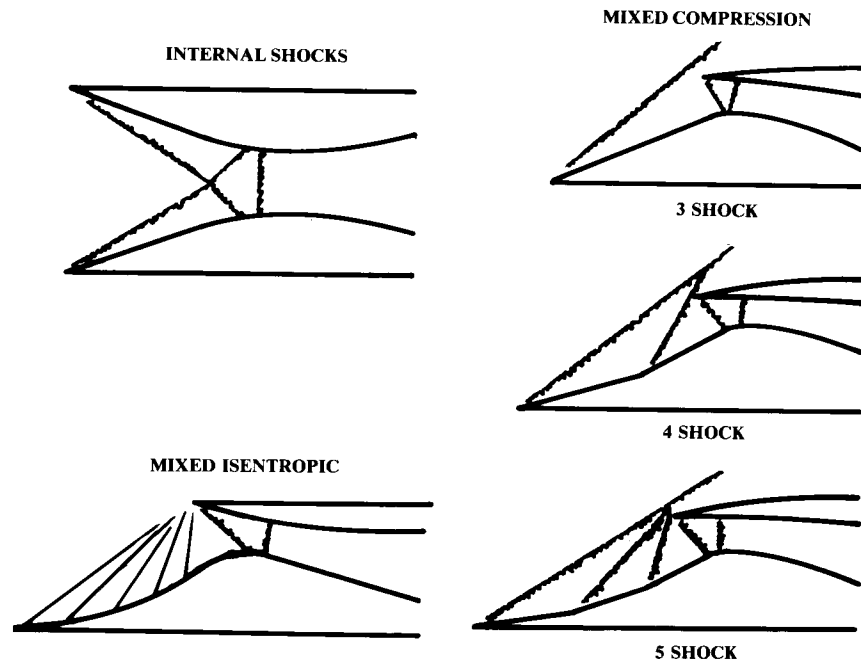


Fig. 10.9 Supersonic inlets—internal and mixed.

The “mixed compression inlet” as shown in Fig. 10.9 uses both external and internal compression to provide high efficiency over a wide Mach number range, with an acceptable amount of external flow turning. Typically one or more external oblique shocks will feed a single internal oblique shock, followed by a final normal shock.

Such an inlet has been used for most aircraft designed to fly above Mach 2.5, including the B-70, which has a 2-D inlet, and the SR-71, which has an axisymmetric inlet. Unstart remains a problem for this type of inlet. Automatically opening doors are used to control unstart.

Mixed-compression inlets are complex, and can be defined only by detailed propulsion analysis beyond the scope of this book. Reference 25 is recommended. The rules of thumb provided above for the dimensions of external-compression inlets give a reasonable first approximation for mixed-compression inlets.

The “diffuser” is the interior portion of an inlet where the subsonic flow is further slowed down to the speed required by the engine. Thus, a diffuser is increasing in cross-sectional area from front to back.

The required length of a diffuser depends upon the application. For a subsonic aircraft such as a commercial transport, the diffuser should be as short as possible without exceeding an internal angle of about 10 deg. Typically, this produces a pitot inlet with a length about equal to its front-face diameter.

For a supersonic application, the theoretical diffuser length for maximum efficiency is about eight times the diameter. Lengths longer than eight times the diameter are permissible but have internal friction losses as well as an additional weight penalty.

A supersonic diffuser shorter than about four times the diameter may produce some internal flow separation, but the weight savings can exceed the engine performance penalty. Diffusers as short as two times the diameter have been used with axisymmetric spike inlets.

For a long diffuser it is important to verify that the cross-sectional area of the flow path is smoothly increasing from the inlet front face back to the engine. This verification is done with a volume-distribution plot of the inlet duct, constructed in the same fashion as the aircraft volume plot shown in Fig. 7.36.

To reduce distortion, some aircraft use a diffuser oversized about 5% that “pinches” the flow down to the engine front-face diameter in a very short distance just before the engine.

Figure 10.10 summarizes the selection criteria for different inlets, based upon design Mach number. Note that these are approximate criteria, and may be overruled by special considerations. Estimated pressure recoveries of these inlets is provided in Chapter 13.

Inlet Location

The inlet location can have almost as great an effect on engine performance as the inlet geometry. If the inlet is located where it can ingest a vortex off the fuselage or a separated wake from a wing, the resulting

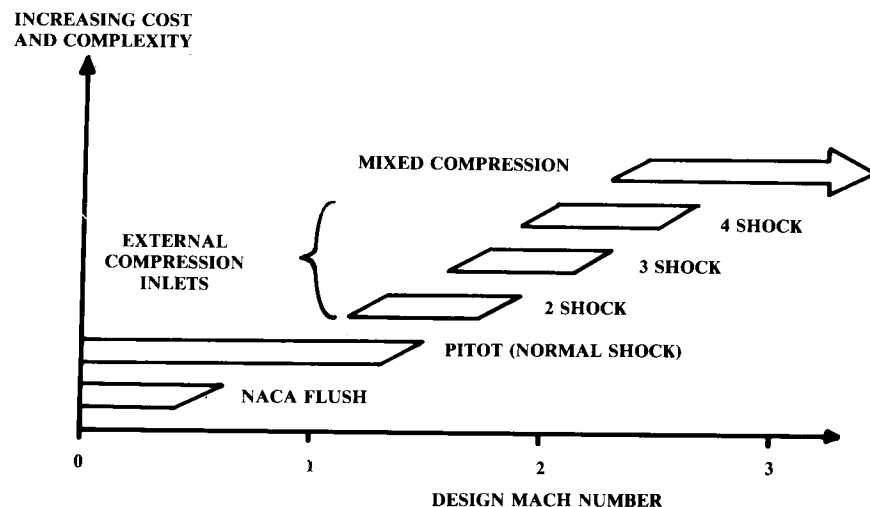


Fig. 10.10 Inlet applicability.

inlet-flow distortion can stall the engine. The F-111 had tremendous problems with its inlets, which were tucked up under the intersection of the wing and fuselage. The A-10 required a fixed slot on the inboard wing leading edge to cure a wake-ingestion problem.

Figures 10.11 and 10.12 illustrate the various options for inlet location. The nose location offers the inlet a completely clean airflow, and was used in most early fighters including the F-86 and MiG 21 as a way of insuring that the fuselage would not cause distortion problems. However, the nose inlet requires a very long internal duct, which is heavy, has high losses, and occupies much of the fuselage volume.

The chin inlet as seen on the F-16 has most of the advantages of the nose inlet but a shorter duct length. The chin inlet is especially good at high angle of attack because the fuselage forebody helps to turn the flow into it.

The location of the nose landing gear is a problem. It cannot be placed forward of the inlet because it would block and distort the flow, and also the nose wheel would tend to throw water and rocks into the inlet. Instead, it is usually placed immediately behind the inlet, which requires that the cowl be deep enough to hold the retracted gear, which can increase cowl drag. Also, the cowl must be strong enough to carry the nose-gear loads.

If two engines are used, twin inlets can be placed in the chin position with the nose wheel located between them. This was used on the North American Rockwell proposal for the F-15, and is seen on the Sukhoi Su-27.

Another problem with the chin inlet is foreign-object ingestion by suction. As a rule of thumb, all inlets should be located a height above the runway equal to at least 80% of the inlet's height if using a low bypass ratio engine, and at least 50% of the inlet's height for a high-bypass-ratio engine.

Side-mounted inlets are now virtually standard for aircraft with twin engines in the fuselage. Side inlets provide short ducts and relatively clean air. Side-mounted inlets can have problems at high angles of attack due to

the vortex shed off the lower corner of the forebody. This is especially severe if the forward fuselage has a fairly square shape.

If side-mounted inlets are used with a single engine, a split duct must be used. Split ducts are prone to a pressure instability that can stall the engine. To minimize this risk, it is best to keep the two halves of the duct separate all the way to the engine front face, although several aircraft have flown with the duct halves rejoined well forward of the engine.

A side inlet at the intersection of the fuselage and a high wing is called an "armpit" inlet. It is risky! The combined boundary layers of the forebody and wing can produce a boundary layer in the wing-fuselage corner that is too thick to remove (boundary-layer removal is discussed later). This type of inlet is especially prone to distortion at angle of attack and sideslip. In many cases however, the armpit inlet does offer a very short internal duct.

An over-fuselage inlet is much like an inverted chin inlet, and has a short duct length but without the problems of nose-wheel location. This was used on the unusual F-107. The upper-fuselage inlet is poor at high angle of attack because the forebody blanks the airflow. Also, many pilots fear that they may be sucked down the inlet if forced to bail out manually.

Placed over the wing and near the fuselage, an inlet encounters problems similar to those of an inverted-armpit inlet. It also suffers at angle of attack.

An inlet above the aft fuselage for a buried engine is used on the L-1011 and B-727, with the inlet located at the root of the vertical tail. This arrangement allows the engine exhaust to be placed at the rear of the fuselage, which tends to reduce fuselage separation and drag. The buried engine with a tail inlet must use an "S-duct." This requires careful design to avoid internal separation. Also, the inlet should be well above the fuselage to avoid ingesting the thick boundary layer.

Inlets set into the wing leading edge can reduce the total aircraft wetted area by eliminating the need for a separate inlet cowl. However, these inlets can disturb the flow over the wing and increase its weight. The wing-root position may also ingest disturbed air off the fuselage.

A podded engine has higher wetted area than a buried engine, but offers substantial advantages that have made it standard for commercial and business jets. Podded engines place the inlet away from the fuselage, providing undisturbed air with a very short inlet duct. Podded engines produce less noise in the cabin because the engine and exhaust are away from the fuselage. Podded engines are usually easier to get to for maintenance. Most are mounted on pylons, but they can also be mounted conformal to the wing or fuselage.

The wing-mounted podded engine is the most commonly used engine installation for jet transports. The engines are accessible from the ground and well away from the cabin. The weight of the engines out along the wing provides a "span-loading" effect, which helps reduce wing weight. The jet exhaust can be directed downward by flaps which greatly increases lift for short takeoff.

On the negative side, the presence of pods and pylons can disturb the airflow on the wing, increasing drag and reducing lift. To minimize this, the pylons should not extend above and around the wing leading edge, as was seen on one early jet transport.

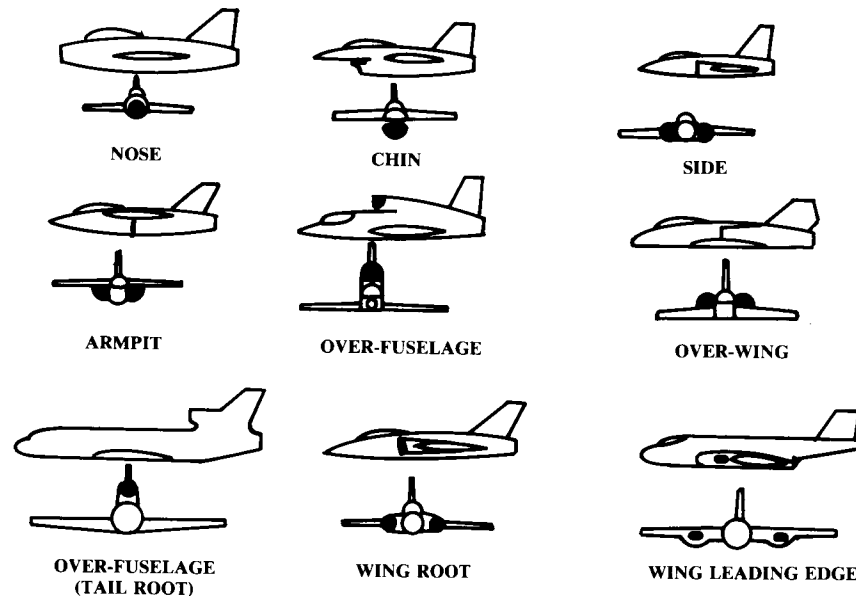


Fig. 10.11 Inlet locations—buried engines.

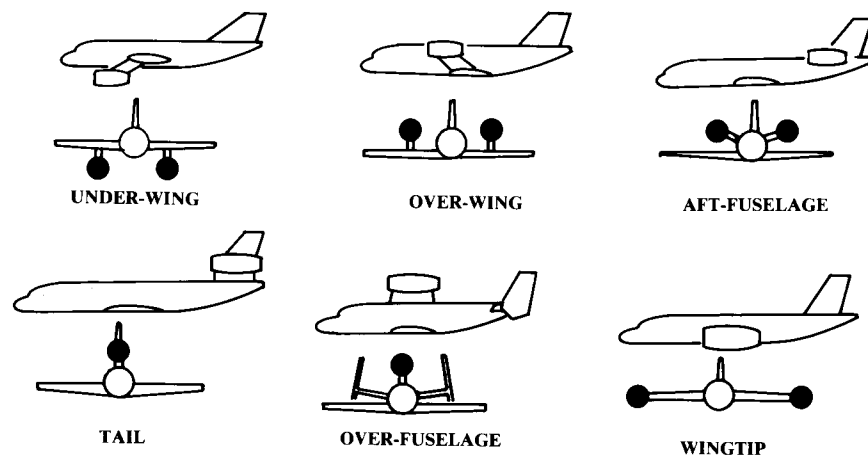


Fig. 10.12 Inlet locations—podded engines.

On the basis of years of wind-tunnel study, design charts for pylon-mounted engines have been prepared that minimize the interference effects of the nacelle pod on the wing. As a classical rule-of-thumb, the inlet for a wing-mounted podded engine should be located approximately two inlet diameters forward and one inlet diameter below the wing leading edge. However, modern computational fluid dynamic (CFD) methods now allow

designing a wing-mounted nacelle much closer to the wing, or even conformal to the wing, without incurring substantial drag increase due to interference. This will be further discussed in Chapter 12.

The wing-mounted nacelle should be angled nose down by about 2–4 deg, and canted nose inward about 2 deg to align it to the local flow under the wing.

To reduce foreign-object ingestion by suction, the inlet of a high-bypass engine should be located about half a diameter above the ground. This requirement increases the required landing-gear height of the under-wing arrangement.

The over-wing podded nacelle reduces the landing-gear height and reduces noise on the ground, but is difficult to get to for maintenance. The inlets can be forward of the wing to minimize distortion, or above it. If an over-wing nacelle is conformal to the wing, the exhaust can be directed over the top of flaps, which, through Coanda effect, turn the flow downward for increased lift.

The other standard engine installation for jet transports is the aft-fuselage mount, usually with a T-tail. This eliminates the wing-interference effects of wing-mounted engines, and allows a short landing gear. However, it increases the cabin noise at the rear of the aircraft.

Also, aft-mounting of the engines tends to move the center of gravity aft, which requires shifting the entire fuselage forward relative to the wing. This shortens the tail moment arm and increases the amount of fuselage forward of the wing, and that necessitates a larger vertical and horizontal tail.

To align the aft nacelle with the local flow, a nose-up pitch of 2–4 deg and a nose outward cant of 2 deg are recommended.

The Illyushin Il-76 uses four aft-podded engines in two twin-engine pods. The L-1011, B-727, and Hawker-Siddeley Trident combine aft-fuselage podded engines with a buried engine using an inlet over the tail.

The DC-10 combines aft-fuselage side pods with a tail-mounted podded engine. This is similar to the tail-mounted inlet for a buried engine, but eliminates the need for an S-duct. However, this arrangement increases the tail weight and doesn't have the fuselage drag-reduction effect. All told, the two installations are probably equivalent.

The supersonic Tupolev Tu-22 ("Blinder") uses twin engines, pod-mounted on the tail, but this arrangement has not been seen on later Soviet supersonic designs.

The over-fuselage podded engine has been used only rarely. Most recently it was used to add a jet engine to the turboprop Rockwell OV-10. Access and cabin noise are undesirable for this installation.

The wing tip-mounted engine has an obvious engine-out controllability problem. It was used on the Soviet supersonic Myasishchev M-52 ("Bouncer"), which also had under-wing engine pods.

Capture-Area Calculation

Figure 10.13 provides a quick method of estimating the required inlet capture area. This statistical method is based upon the design Mach number and the engine mass flow in pounds per second.

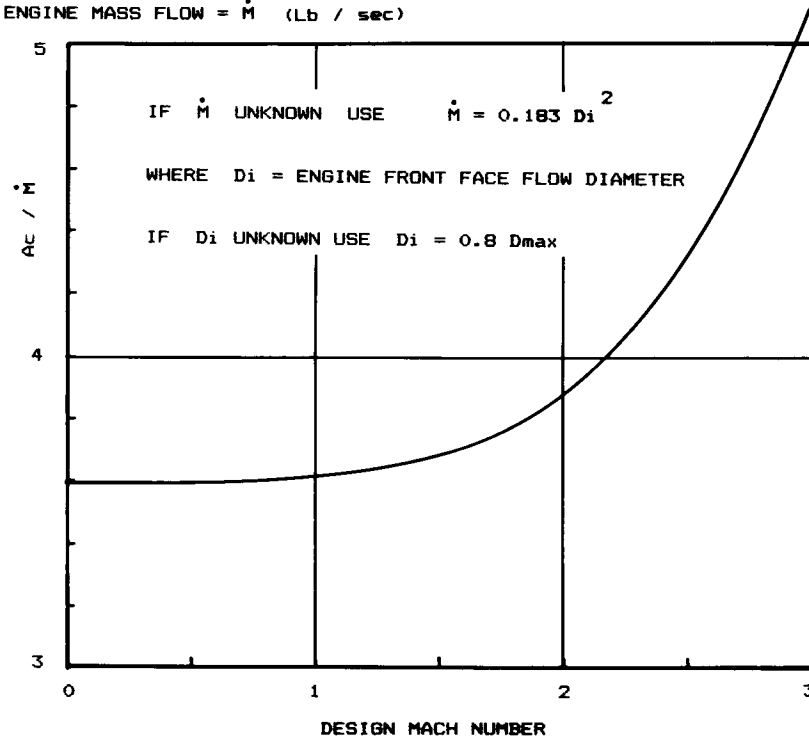
INLET CAPTURE AREA = A_c (square inches)ENGINE MASS FLOW = $\dot{M}$ (Lb / sec)

Fig. 10.13 Preliminary capture area sizing.

If mass flow is not known, it may be estimated as 0.18 times the square of the engine front-face diameter in inches, or as 0.12 times the square of the maximum engine diameter. To determine the required capture area, the mass flow is multiplied by the value from Fig. 10.13.

This capture-area estimation is adequate for initial layout and rough analysis, but not accurate enough for a good configuration layout. A better estimate of capture area should be made during configuration layout based upon the actual mass flow of the engine, as described below.

In a jet propulsion system, the engine is the boss. It takes the amount of air it wants, not what the inlet wants to give it. If the inlet is providing more air than the engine wants, the inlet must spill the excess out the front. If the inlet is not providing what the engine needs, it will attempt to suck in the extra air required.

The inlet capture area must be sized to provide sufficient air to the engine at all aircraft speeds. For many aircraft the capture area must also provide "secondary air" for cooling and environmental control, and also provide for the air bled off the inlet ramps to prevent boundary-layer buildup.

Figure 10.14 defines the capture area for a subsonic inlet. A typical subsonic jet inlet is sized for cruise at about Mach 0.8–0.9, and the inlet must

slow the air to about Mach 0.4 for most engines. Since this is subsonic flow, the inlet does not need to do all the work itself. As shown in Fig. 10.14, the expansion associated with slowing the flow from its freestream velocity at "infinity" takes place about half within and half outside the inlet duct.

The area at the inlet front face is both the capture area and the throat area. It can be calculated from the following isentropic compressible flow relationship:

$$\frac{A_{throat}}{A_{engine}} = \frac{(A/A^*)_{throat}}{(A/A^*)_{engine}} \quad (10.16)$$

$$\frac{A}{A^*} = \frac{1}{M} \left(\frac{1 + 0.2M^2}{1.2} \right)^3 \quad (10.17)$$

where A^* is the area of the same flow at sonic speed.

For a typical inlet designed to a cruise speed of Mach 0.8, the inlet must slow the air from about Mach 0.6 down to Mach 0.4. The air is slowed from Mach 0.8–0.6 outside the inlet.

Equations (10.16) and (10.17) give the ratio between throat area and engine front-face area as 1.188/1.59, or 0.75 (for this example). Taking the square root gives a diameter ratio of about 0.88, which is reasonable. Note that a subsonic inlet generally does not require bleed air, since secondary air is obtained from separate, small NACA flush inlets in most subsonic aircraft.

Equations (10.16) and (10.17) may be used to determine the capture area for a supersonic pitot inlet with negligible bleed or secondary airflow by finding the Mach number behind the normal shock from Appendix A.2. For other supersonic inlets the required capture area must be determined by considering the air-flow requirements for the engine, bleed, and secondary

In sizing a supersonic inlet, a variety of flight conditions must be considered to find the largest required capture area. Typically this will be at the aircraft maximum Mach number, but may also occur during takeoff or subsonic cruise. If the maximum required capture-area occurs during take-

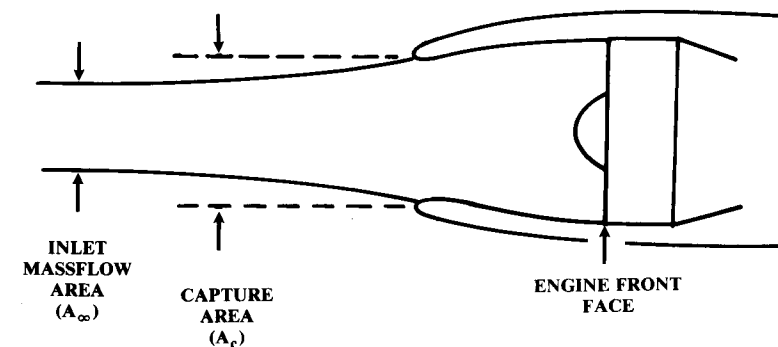


Fig. 10.14 Subsonic inlet capture area.

off, consider using auxiliary “suck-in” doors during takeoff. This allows the inlet to be sized to another, lesser capture area requirement.

To avoid the confusion of differing velocities and densities, the airflow is defined by mass flow in pounds mass per second. Mass flow is related to flow conditions by Eq. (10.18), where g is 32.2 ft/s².

$$\dot{m} = g \rho V A, \text{ lb/s}$$

(10.18)

Figure 10.15 defines the capture area geometry for a supersonic ramp or cone inlet. This inlet is shown at the design case, which is known as “shock-on-cowl.” At this Mach number and ramp angle, the initial oblique shock is almost touching the cowl lip. If the auxiliary doors are shut and the shock is on cowl, the geometric capture area is providing exactly the right amount of air for the engine, bleed, and secondary flow.

Usually the inlet ramp geometry provides shock-on-cowl at about Mach 0.1–0.2 above the aircraft’s maximum speed, giving a safety margin for speed overshoot and engine mass-flow fluctuations.

If the total mass flow required by the engine, bleed, and secondary flow is known, then Eq. (10.18) can be solved for the required cross-sectional area upstream of the inlet (at “infinity”) using the freestream values for density and velocity. This calculated area is identical to the capture area in the design case (shock-on-cowl) since all of the air in the capture area is going into the inlet.

The required engine mass flow is provided by the engine manufacturer, and is a function of the Mach number, altitude, and throttle setting (percent power). Usually the manufacturer’s data should be increased by 3% to allow for manufacturing tolerances.

The secondary airflow requirements are accurately determined by an evaluation of the aircraft’s subsystems such as environmental control. For initial capture-area estimation, Table 10.2 (from Ref. 26) provides secondary airflow as a fraction of engine mass flow.

Table 10.2 Secondary airflow (typical)²⁶

	$\dot{m}_s/\dot{m}_e$
Engine	
Nacelle cooling	0–0.04
Oil cooling	0–0.01
Ejector nozzle air	0.04–0.20
Hydraulic system cooling	0–0.01
Environmental control system	0.02–0.05
cooling air (if taken from inlet)	
Typical totals	
Fighter	0.20
Transport	0.03

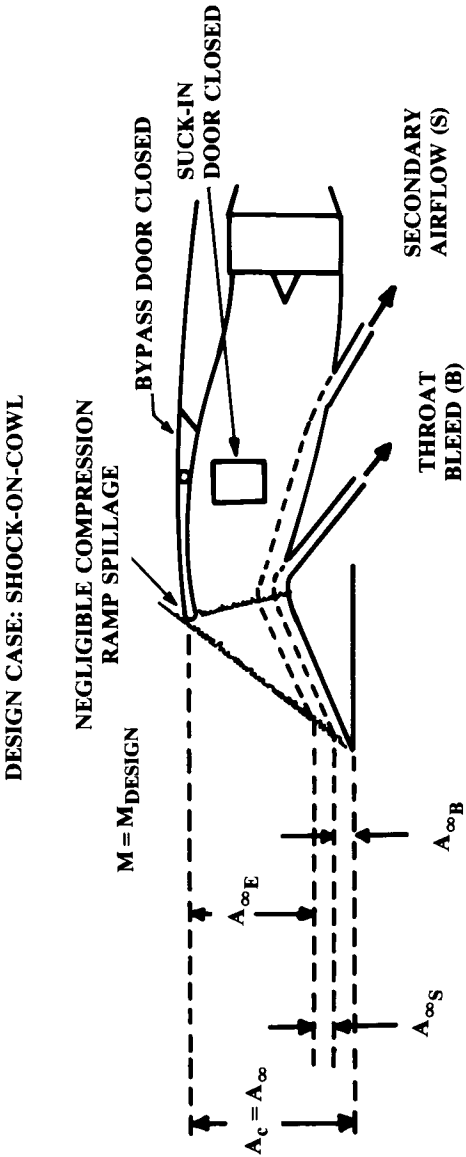


Fig. 10.15 Supersonic inlet capture area—on design.

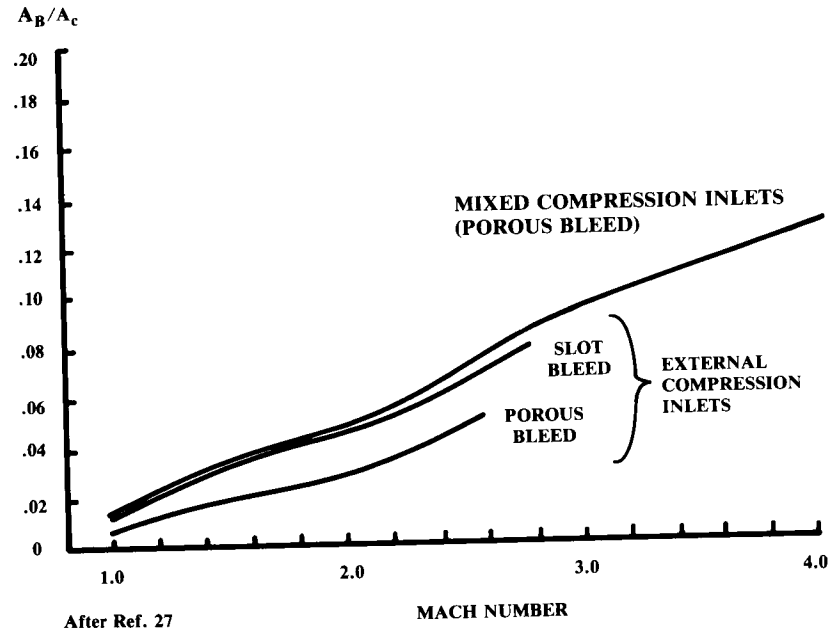


Fig. 10.16 Typical boundary layer bleed area.

Inlet boundary layer bleed should also be determined analytically, but can be approximated using Fig. 10.16, taken from Ref. 27. This estimates the required extra capture area for bleed as a percent of the capture area required for the engine and secondary airflow.

The capture area is therefore determined as in Eq. (10.19), using Table 10.2 and Fig. 10.16.

$$A_{\text{capture}} = \left[\frac{\dot{m}_E(1 + \dot{m}_S/\dot{m}_E)}{g\rho_\infty V_\infty} \right] \left(1 + \frac{A_B}{A_c} \right) \quad (10.19)$$

Figure 10.15 shows the inlet operating at its design condition, shock-on-cowl, where the geometric capture area equals the freestream area of the air actually taken into the inlet and used. If the freestream Mach number is reduced, the oblique shock angle drops, which moves the oblique shock in front of the cowl, as shown in Fig. 10.17a.

Since the airflow is parallel to the ramp, it can be seen that the freestream cross-sectional area of the air that actually goes into the inlet has been reduced. Part of the air defined by the geometric capture area is now spilled after being compressed. This represents wasted work and increased drag compared to the case of shock-on-cowl.

If the mass-flow demand exactly equals the mass flow shown going into the inlet in Fig. 10.17a (i.e., capture area less compression-ramp spillage), then the engine and inlet duct are still “matched” and the normal shock will be at the cowl lip, as shown in Fig. 10.17a.

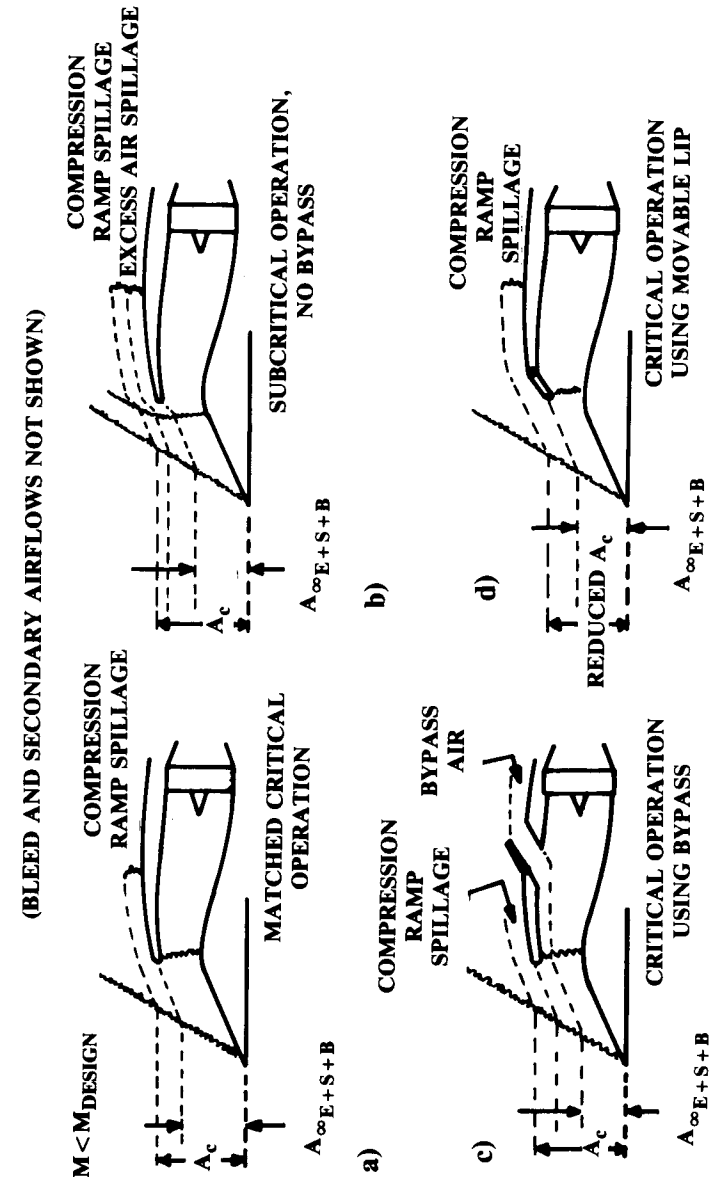


Fig. 10.17 Off-design inlet operation.

However, the engine demand is usually reduced at a slower speed. The excess air is simply rejected by the inlet, as shown in Fig. 10.17b. (Remember, the engine is the boss!) This pushes the normal shock forward of the inlet and creates a much larger spillage drag than for the matched condition.

Two approaches to move the normal shock back to the cowl lip are shown in Figs. 10.17c and 10.17d. By opening a bypass door in the diffuser section, the excess air can be taken into the inlet and thrown away before reaching the engine. While an inlet bypass will create some additional drag, the total is reduced compared with the case in Fig. 10.17b.

(Do not confuse inlet bypass air with the engine bypass air. Inlet bypass air is dumped out of the inlet before it reaches the engine, and is therefore not a contributor to thrust. Engine bypass air is exited after being accelerated by the compressor, and does contribute to thrust.)

Another approach for returning the normal shock to the inlet lip is to move the cowl lip down, reducing the capture area as shown in Fig. 10.17d. This is complex and heavy to mechanize, and virtually impossible for an axisymmetric inlet.

It is also possible to translate the ramp or spike fore and aft to maintain shock-on-cowl at different Mach numbers. This is often employed with high-speed spike inlets.

The ratio between the air flow actually going into the inlet and the total possible air flow (i.e., the airflow of the capture area) is called the "capture area ratio," or "inlet mass flow ratio." The total mass flow actually going into the inlet is the mass flow required for the engine plus secondary airflow plus bleed airflow plus inlet bypass air, if any.

Capture-area ratio is calculated by determining the required mass flow and dividing by the mass flow through the capture area far upstream (Eq. (10.20). Note that capture-area ratio is generally critical for conditions in which the inlet bypass doors are closed (no bypass mass flow).

$$\frac{A_\infty}{A_c} = \frac{\dot{m}_E + \dot{m}_S + \dot{m}_{BL} + \dot{m}_{bypass}}{g \rho_\infty V_\infty A_c} \quad (10.20)$$

The capture-area ratio in subsonic flow can be greater than, equal to, or less than 1. In supersonic flow, it can only be equal to or less than 1.

Boundary-Layer Diverter

Any object moving through air will build up a boundary layer on its surface. In the last section, boundary-layer bleed was included in the capture-area calculation. This boundary-layer bleed was used to remove the low-energy boundary layer air from the compression ramps, to prevent shock-induced separation.

The aircraft's forebody builds up its own boundary layer. If this low-energy, turbulent air is allowed to enter the engine, it can reduce engine performance subsonically and prevent proper inlet operation supersonically. Unless the aircraft's inlets are very near the nose (within two to four inlet diameters), some form of boundary-layer removal should be used just in front of the inlet.

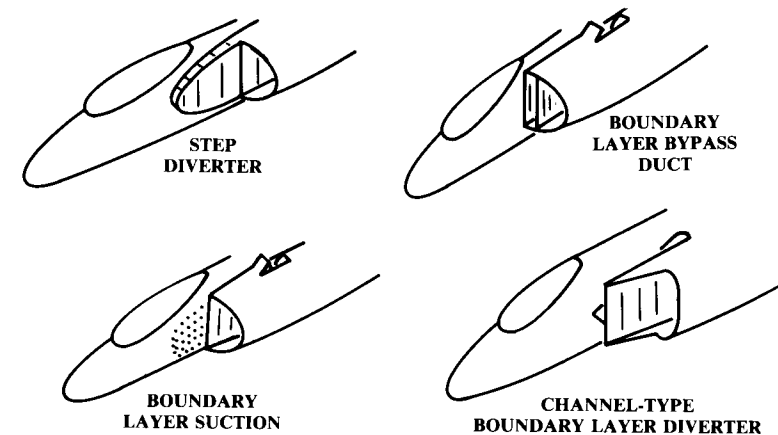


Fig. 10.18 Boundary layer removal.

The four major varieties of boundary-layer diverter are shown in Fig. 10.18. The step diverter is suitable only for subsonic aircraft, and relies upon the boundary layer itself for operation. The boundary layer consists of low-energy air, compared to the air outside of the boundary layer.

The step diverter works by forcing the boundary-layer air to either climb the step, pushing aside high-energy air outside the boundary layer, or to follow the step, pushing aside other boundary-layer air which is of lower energy. If the step diverter is properly shaped, the latter option prevails.

The step diverter should have an airfoil-like shape that is faired smoothly to the nacelle. The diverter should extend about one inlet diameter forward of the inlet, and should be have a depth equal to roughly 2–4% of the forebody length ahead of the inlet.

The boundary-layer bypass duct (simply a separate inlet duct) admits the boundary-layer air and ducts it to an aft-facing hole. The internal duct shape should expand roughly 30% from intake to exit to compensate for the internal friction losses.

The suction form of boundary-layer diverter is similar. The boundary-layer air is removed by suction through holes or slots just forward of the inlet and ducted to an aft-facing hole. This type of diverter does not benefit from the ram impact of the boundary-layer air, and therefore does not work as well.

The channel diverter (Fig. 10.19) is the most common boundary-layer diverter for supersonic aircraft. It provides the best performance and the least weight in most cases. The inlet front face is located some distance away from the fuselage, with a "splitter plate" to insure that the boundary-layer air does not get into the inlet. The boundary-layer air is caught between the splitter plate and the fuselage, and pushed out of the resulting channel by the diverter ramps. The diverter ramps should have an angle of no more than about 30 deg.

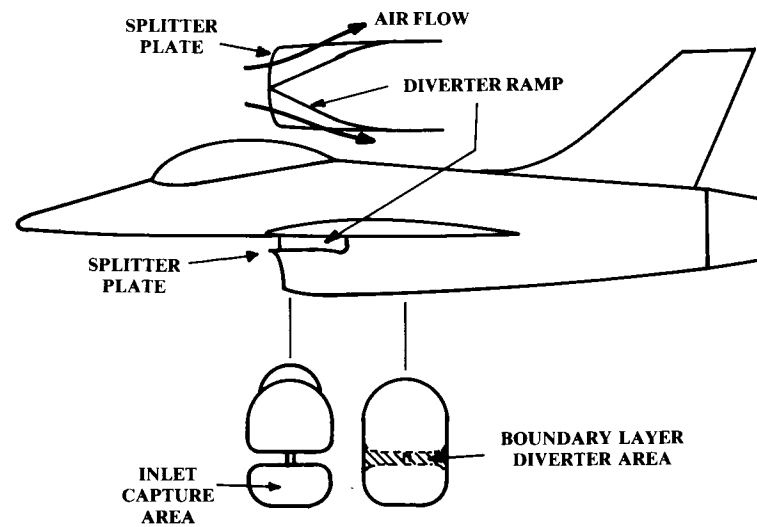


Fig. 10.19 Boundary layer diverter.

The required depth of a boundary-layer diverter depends on the depth of the boundary layer itself. This cannot be simply calculated. The classic boundary-layer equations assume a flat plate, which is unlike a fuselage forebody. The 3-D effects of a real forebody tend to reduce boundary-layer buildup compared to a flat plate.

A very good rule of thumb for the required thickness of a boundary-layer diverter is that it should be between 1 and 3% of the fuselage length in front of the inlet, with the larger number for fighters which go to high angle of attack.

As will be discussed in Chapter 12, the drag of a boundary-layer diverter depends upon its frontal area. During conceptual layout, the fuselage and inlet should be designed to minimize this area, shown shaded in Fig. 10.19.

Nozzle Integration

The fundamental problem in jet engine nozzle design is the mismatch in desired exit areas at different speeds, altitudes, and thrust settings. The engine can be viewed as a producer of high-pressure subsonic gases. The nozzle accelerates those gases to the desired exit speed, which is controlled by the exit area.

The nozzle must converge to accelerate the exhaust gases to a high subsonic exit speed. If the desired exit speed is supersonic, a converging-diverging nozzle is required.

The exit area to obtain a desired exhaust velocity depends upon the engine mass flow (i.e., percent power). This is especially a problem with afterburning engines, in which the desired exit area for supersonic afterburning operation can be three times the desired area for subsonic, part-thrust operation.

Typical nozzles are shown in Fig. 10.20. In the past, the nozzle of a jet engine was considered an integral part of the engine, to be installed on the aircraft without question or change. This is still the case for subsonic commercial aircraft, but is changing for supersonic military aircraft due to the emergence of 2-D and other advanced nozzles.

The fixed convergent nozzle is almost universally used for subsonic commercial turbojet and turbofan engines. The nozzle exit area is selected for cruise efficiency, resulting in a loss of theoretical performance at lower speeds. However, the gain in simplicity and weight reduction of the fixed nozzle more than makes up for the performance loss in most subsonic applications.

For an aircraft which occasionally flies at high-subsonic to low-supersonic speeds, a variable-area convergent nozzle allows a better match between low-speed, part-thrust operation and the maximum speed and thrust conditions. The nozzle shown has a fixed outer surface, which causes a "base" area when the nozzle inside is in the closed position.

Such a nozzle was used on many early transonic fighters, but is not typically used today. Instead, the convergent-iris nozzle is used to vary the area of a convergent nozzle without introducing a base area.

Another means to vary the exit area of a convergent nozzle is the translating plug. This was used on the engine for the Me-262, the first jet to be employed in combat in substantial numbers. The plug slides aft to decrease exit area.

The ejector nozzle takes engine bypass air that has been used to cool the afterburner and ejects it into the exhaust air, thus cooling the nozzle as well. The variable-geometry convergent-divergent ejector nozzle is most commonly applied to supersonic jet aircraft. It allows varying the throat and

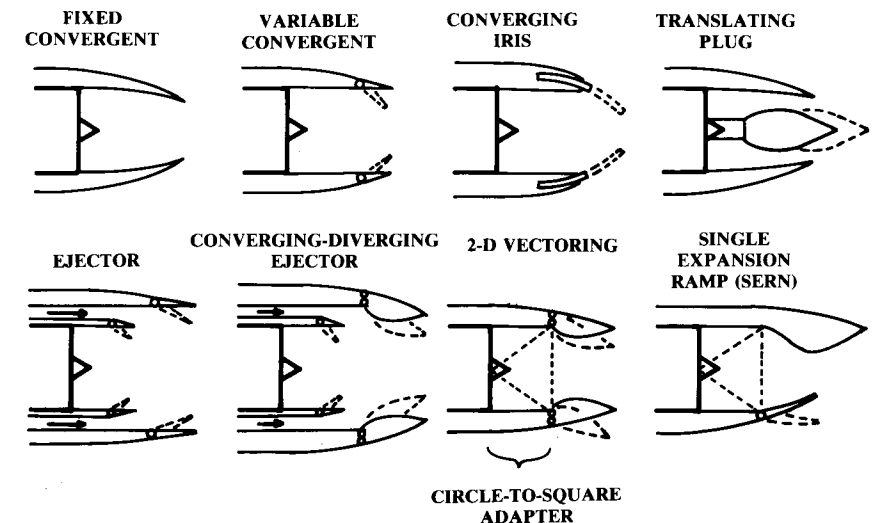


Fig. 10.20 Types of nozzles.

exit areas separately for maximum engine performance throughout the flight envelope.

If an existing engine is used in the design, or if a hypothetical engine data package has been obtained from an engine company, the nozzle areas will be provided for the design flight regime. If not, the nozzle areas must be estimated because they have a substantial effect upon the calculated aircraft wave drag and boattail drag.

For initial design layout, a reasonable approximation can be made based upon the estimated capture area. For a subsonic convergent nozzle or a convergent-divergent nozzle in the closed position, the required exit area is approximately 0.5–0.7 times the capture area. For maximum supersonic afterburning operation, the required exit area is about 1.2–1.6 times the capture area.

As mentioned, nozzle arrangement can have a substantial effect on boat-tail drag. This is the drag due to separation on the outside of the nozzle and aft fuselage. To reduce boattail drag to acceptable levels, the closure angles on the aft fuselage should be kept below 15 deg, and the angles outside of the nozzle should be kept below 20 deg in the nozzle-closed position.

Jet engines mounted next to each other produce an interference effect that reduces net thrust. To minimize this, the nozzles should be separated by about one to two times their maximum exit diameter. The area between them should taper down like the back of an airfoil, terminating just before the nozzles. However, this arrangement increases weight and wetted area so many fighters have twin engines mounted right next to each other despite the increased interference.

10.4 PROPELLER-ENGINE INTEGRATION

Propeller Sizing

The actual details of the propeller design such as the blade shape and twist are not required to lay out a propeller-engine aircraft. But the diameter of the propeller, the dimensions of the engine, and the required inlets and exhausts must be determined.

Generally speaking, the larger the propeller diameter, the more efficient the propeller will be. The old rule of thumb was “keep it as long as possible, as long as possible.” The limitation on length is the propeller tip speed, which should be kept below sonic speed.

The tip of a propeller follows a helical path through the air. Tip speed is the vector sum of the rotational speed [Eq. (10.21)] and the aircraft’s forward speed as defined in Eq. (10.22).

$$(V_{\text{tip}})_{\text{static}} = \pi n d / 60 \quad (\text{ft/s}) \quad (10.21)$$

where

n = rotational rate (rpm) obtained from engine data

d = diameter

$$(V_{\text{tip}})_{\text{helical}} = \sqrt{V_{\text{tip}}^2 + V^2} \quad (10.22)$$

Note: watch the units!

At sea level the helical tip speed of a metal propeller should not exceed 950 fps. A wooden propeller, which must be thicker, should be kept below 850 fps. If noise is of concern, the upper limit for metal or wood should be about 700 fps during takeoff.

Equations (10.23–10.25) provide an estimate of the propeller diameter as a function of horsepower (Ref. 28). The propeller diameters obtained from these equations should be compared to the maximum diameters obtained from tip-speed considerations, and the smaller of the two values used for initial layout.

$$\text{Two blade: } d = 22 \sqrt[4]{\text{Hp}} \quad (10.23)$$

$$\text{Three blade: } d = 18 \sqrt[4]{\text{Hp}} \quad (10.24)$$

$$\text{Three blade (agricultural): } d = 20 \sqrt[4]{\text{Hp}} \quad (10.25)$$

As forward velocity increases the angle of attack seen by the blades of a fixed-pitch propeller will decrease. This limits the thrust obtained at higher speeds. If the fixed pitch is increased, the blades will tend to stall at low speeds, which reduces low-speed thrust. A fixed-pitch propeller is called a “cruise prop” or “climb prop” depending upon the flight regime the designer has decided to emphasize.

A variable-pitch propeller can be used to improve thrust across a broad speed range. A “controllable-pitch” propeller has its pitch directly controlled by the pilot through a lever alongside the throttle. A “constant-speed” propeller is automatically controlled in pitch to maintain the engine at its optimal RPM.

Most aircraft propellers have a “spinner,” a cone- or bullet-shaped fairing at the hub. The inner part of the propeller contributes very little to the thrust. A spinner pushes the air out to where the propeller is more efficient. Also, a spinner streamlines the nacelle. Ideally, the spinner should cover the propeller out to about 25% of the radius, although most spinners are not that large.

To further streamline the nacelle, some aircraft designers use a “prop extension,” a short shaft which locates the propeller 2–4 in. farther forward (or aft) of the engine. If the propeller is located much farther away from the engine, a complicated drive shaft with a separate bearing support for the propeller must be used. This type of installation was used in the P-39, which had a piston engine behind the cockpit and a drive shaft to the forward-mounted propeller. Similarly, the BD-5 had a drive shaft to a rear-mounted pusher propeller.

Propeller Location

A matrix of possible propeller locations is shown in Fig. 10.21. A tractor installation has the propeller in front of its attachment point (usually the motor). A pusher location has the propeller behind the attachment point.

The Wright Flyer was a pusher. However, the tractor location has been standard for most of the history of aviation. The conventional tractor loca-

tion puts the heavy engine up front, which tends to shorten the forebody, allowing a smaller tail area and improved stability. The tractor location also provides a ready source of cooling air, and places the propeller in undisturbed air.

The pusher location is now seeing wider use because of its advantages. Most importantly, it can reduce aircraft skin friction drag because the pusher location allows the aircraft to fly in undisturbed air. With a tractor propeller the aircraft flies in the turbulence from the propeller wake.

The fuselage-mounted pusher propeller can allow a reduction in aircraft wetted area by shortening the fuselage. The inflow caused by the propeller allows a much steeper fuselage closure angle without flow separation than otherwise possible. The canard-pushers combination is especially favorable because the canard requires a shorter tail arm than the aft tail.

The pusher propeller reduces cabin noise because the engine exhaust is pointed away from the cabin, and because the windscreen isn't buffeted by propwash. Also, the pusher arrangement usually improves the pilot's outside vision.

The pusher propeller may require longer landing gear because the aft location causes the propeller to dip closer to the runway as the nose is lifted for takeoff. The propeller should have at least 9 in. of clearance in all attitudes.

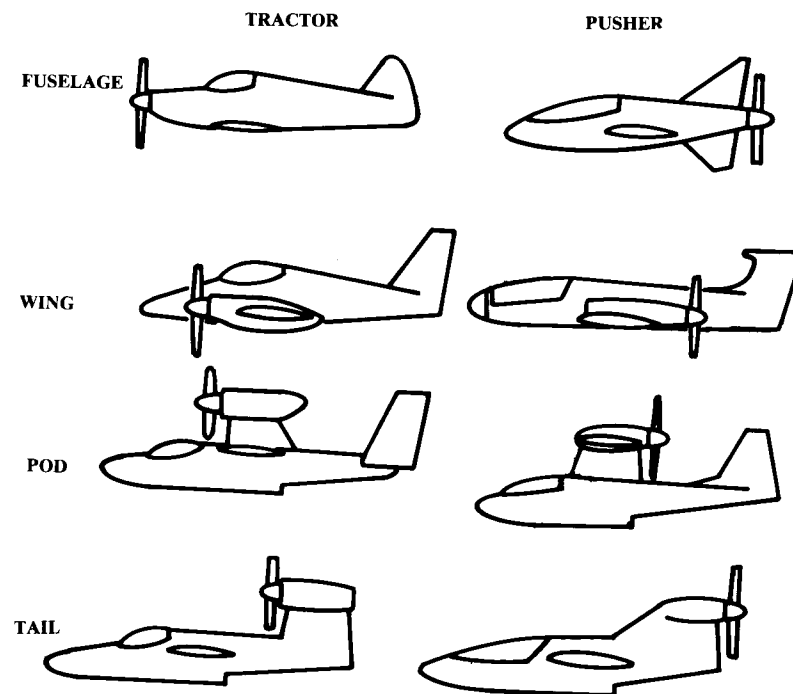


Fig. 10.21 Propeller location matrix.

The pusher-propeller is also more likely to be damaged by rocks thrown up by the wheels. A pusher location for a turboprop propeller can create problems due to the engine exhaust impinging upon the propeller.

The Cessna Skymaster and Rutan Defiant use a combination of pusher and tractor engines on the fuselage.

Wing mounting of the engines is normally used for multiengine designs. Wing mounting of engines reduces wing structural weight through a span-loading effect, and reduces fuselage drag by removing the fuselage from the propeller wake.

Wing mounting of engines introduces engine-out controllability problems that force an increase in the size of the rudder and vertical tail. Also, care must be taken to insure that the crew compartment is not located within plus or minus 5 deg of the propeller disk, in case a blade is thrown through the fuselage.

Most twin-engine aircraft are of low-wing design. For these, the location of the engine and propeller on the wing requires a longer landing gear. Frequently the propeller will be raised above the plane of the wing to reduce landing-gear height. This causes additional interference between the wing and propeller.

The wing-mounted pusher arrangement has been seen on the Beech Starship and B-36. This arrangement tends to lengthen the forebody and require a very long landing gear.

Also, the propeller is half in the wake from under the wing and half in the wake from over the wing. The pressure differences between these two wakes can cause the propeller to lose efficiency and produce vibrations. This is minimized by locating the propeller as far as possible behind the wing.

Upper fuselage pods and tail-mounted pods tend to be used only for seaplane and amphibian designs, which need a huge clearance between the water and the propeller (minimum of 18 in., preferably one propeller diameter). The high thrust line can cause undesirable control characteristics in which application of power for an emergency go-around produces a nose-down pitching moment.

Engine-Size Estimation

The required horsepower has previously been calculated. The dimensions of an engine producing this power must now be determined. In propeller aircraft design it is far more common to size the aircraft to a known, fixed-size engine as opposed to the rubber-engine aircraft sizing more common in jet-aircraft design.

In fact, most propeller-aircraft designs are based around some production engine, probably because very few new piston or turboprop engines have been designed and certified. Most piston engines in production were designed three decades ago. The high cost of developing and certifying a new engine, and the relatively small market, prevent new engines from appearing.

However, rubber-engine trade studies can point to the optimal existing engine. Also, the use of rubber-engine trade studies for comparison of alternate technologies (such as composite vs aluminum structure) can prevent a bias in the results due to the use of a fixed engine size.

Table 10.3 Scaling laws for piston and turboprop engines

$$X_{\text{scaled}} = X_{\text{actual}} SF^b; b \text{ from table values}$$

$$SF = \text{bhp}_{\text{scaled}} / \text{bhp}_{\text{actual}}$$

X	Piston engines			Turboprop
	Opposed	In-line	Radial	
Weight	0.78	0.78	0.809	0.803
Length	0.424	4.24	0.310	3.730
Diameter	*	*	0.130	0.120

*Width and height vary insignificantly within $\pm 50\%$ horsepower.

Table 10.4 Piston and turboprop statistical models

$$X = a(\text{bhp})^b \text{ (lb or in.)}$$

X	Piston engines						Turboprop	
	Opposed		In-line		Radial		a	b
	a	b	a	b	a	b		
Weight	5.47	0.780	5.22	0.780	4.90	0.809	1.67	0.803
Length	3.86	0.424	5.83	0.424	6.27	0.310	4.14	0.373
Diameter	Width 32–34 in. Height 22–25 in.		Width 17–19 in. Height 24–26 in.		20.2	0.130	9.48	0.120
Typical propeller rpm	2770		2770		2300			
Applicable bhp range	60–500		100–300		200–2000		400–5000	

If a production engine is to be used, dimensional and installation data can be obtained from the manufacturer. If a rubber-engine is to be used, an existing engine can be scaled using the scaling equations defined in Table 10.3. Alternatively, the statistical models defined in Table 10.4 can be used to define a nominal engine. The equations in Tables 10.3 and 10.4 were developed by the author from data taken from Ref. 1.

Tables 10.3 and 10.4 include equations for four different types of propeller powerplant. The horizontally-opposed piston engine sees most use today. In-line and radial engines were common up to the 1950's, but are rare today in the Western countries. In Soviet-block countries large radial engines are still in production for agricultural and utility aircraft. The radial arrangement provides better piston cooling for a high-horsepower piston engine.

Piston-Engine Installation

Piston engines have special installation requirements that can greatly affect the configuration layout. These are illustrated in Fig. 10.22.

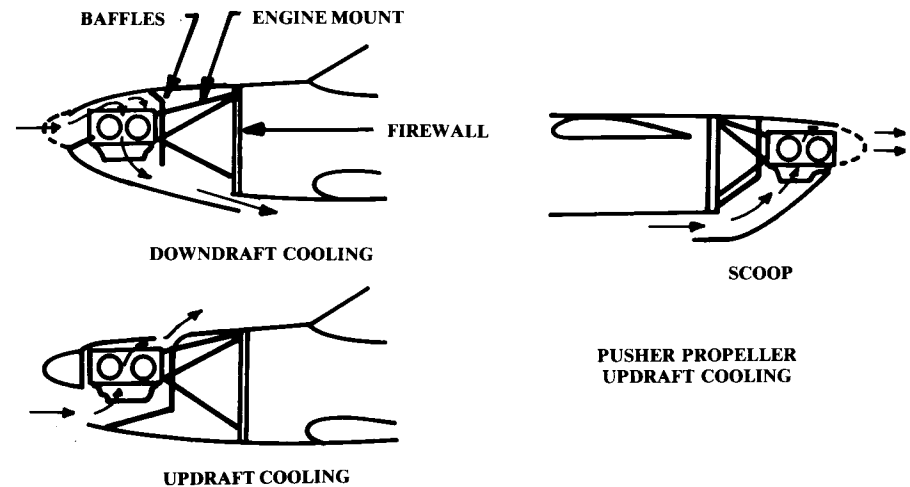
Cooling is a major concern. Up to 10% of the engine's horsepower can be wasted by the drag associated with taking in cooling air, passing it over the engine, and exiting it.

To minimize this cooling drag, the cooling-air mass flow should be kept as small as possible and used as efficiently as possible. As a rough rule of thumb, the cooling-air intake should be about 30–50% of the engine frontal area. The exit should be about 30% larger, and may be variable in area ("cowl flaps") to better control cooling airflow.

For tractor engines, the cooling-air intake is usually located directly in front of the engine cylinders. The air is diverted over the top of the engine by "baffles," which are flat sheets of metal that direct the airflow within the engine compartment. The air then flows down through and around the cylinders into the area beneath the engine, and then exits through an aft-facing hole below the fuselage. This is referred to as "down-draft" cooling.

Down-draft cooling exits the air beneath the fuselage, which is a high-pressure area and therefore a poor place to exit air. "Up-draft" cooling flows the cooling air upwards through the cylinders and exits it into low-pressure air above the fuselage, creating more efficient cooling flow due to a suction effect.

However, updraft cooling dumps hot air in front of the windscreen; this can heat up the cabin. An engine oil leak can coat the windscreen with black oil. Aircraft engines have the exhaust pipes below the cylinders, so updraft cooling causes the cooling air to be heated by the exhaust pipes before reaching the cylinders.

**Fig. 10.22** Piston engine installation.

For pusher engines cooling is much more difficult. On the ground a front-mounted propeller blows air into the cooling intakes. This is not the case for a pusher engine. Also, the cooling-air intakes for a pusher engine are at the rear of the fuselage where the boundary layer is thick and slow-moving. For these reasons virtually all piston-pushers use updraft cooling with a large scoop mounted below the fuselage. Also, internal fans are sometimes used to improve cooling on pusher configurations.

Figure 10.22 also shows the motor mount and firewall. The motor mount—usually fabricated from welded steel tubing—transfers the engine loads to the corners of the fuselage or the longerons. Typically the motor mount extends the engine forward of the firewall by about half the length of the engine. This extra space is used for location of the battery and nosewheel steering linkages.

The firewall is typically a 0.015-in. steel sheet (stainless or galvanized) attached to the first structural bulkhead of the fuselage or nacelle. Its purpose is to prevent a fire in the engine compartment from damaging the aircraft structure or spreading into the rest of the aircraft.

The firewall should not be broken with cutouts (such as for a retractable nose wheel). All controls, hoses, and wires that pass through the firewall have to be sealed with fireproof fittings.

Piston-engine installation is covered in depth in Ref. 29.

10.5 FUEL SYSTEM

An aircraft fuel system includes the fuel tanks, fuel lines, fuel pumps, vents, and fuel-management controls. Usually the tanks themselves are the only components that impact the overall aircraft layout, although the winglets on the round-the-world Rutan Voyager were added solely to raise the fuel vents above the wing tanks when the wing tips bent down to the runway on takeoff.

There are three types of fuel tank: discrete, bladder, and integral. Discrete tanks are fuel containers which are separately fabricated and mounted in the aircraft by bolts or straps. Discrete tanks are normally used only for small general aviation and homebuilt aircraft. Discrete tanks are usually shaped like the front of an airfoil and placed at the inboard wing leading edge, or are placed in the fuselage directly behind the engine and above the pilot's feet.

Bladder tanks are made by stuffing a shaped rubber bag into a cavity in the structure. The rubber bag is thick, causing the loss of about 10% of the available fuel volume. However, bladders are widely used because they can be made "self-sealing." If a bullet passes through a self-sealing tank, the rubber will fill in the hole preventing a large fuel loss and fire hazard. This offers a major improvement in aircraft survivability as approximately a third of combat losses are attributed to hits in the fuel tanks.

Integral tanks are cavities within the airframe structure that are sealed to form a fuel tank. Ideally, an integral tank would be created simply by sealing existing structure such as wing boxes and cavities created between two fuselage bulkheads.

Despite years of research, integral tanks are still prone to leaks as witnessed upon the introduction of the B-1B into service. Due to the fire hazard in the event of a leak or battle damage, integral tanks should not be used near personnel compartments, inlet ducts, gun bays, or engines.

The fire hazard of an integral tank can be reduced by filling the tank with a porous foam material, but some fuel volume is lost. Approximately 2½% of the fuel volume is displaced by the foam. In addition, another 2½% of the volume is lost because the foam tends to absorb fuel. This increases the unusable fuel weight. Furthermore, the foam itself weighs roughly 1.3 lb per cubic ft.

The required volume of the fuel tanks is based upon the total required fuel, as calculated during the mission sizing. Densities for various fuels are provided in Table 10.5. The lower values represent hot-day densities. The higher densities are the 32°F values. The actual fuel volume required can be calculated using the fact that one gallon occupies 231 cubic in.

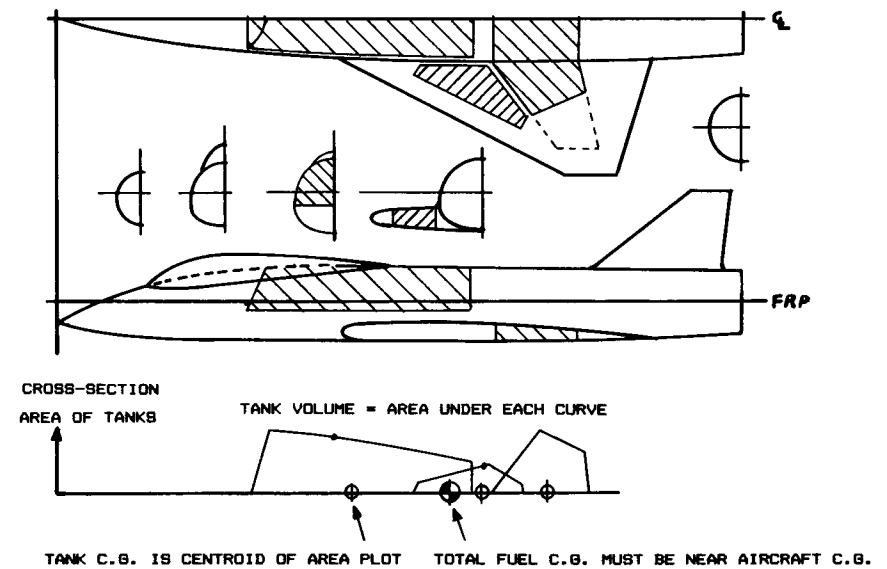


Fig. 10.23 Fuel tank volume plotting.

Table 10.5 Fuel densities (lb/gal)

	Average actual density		Mil-spec density
	0°F	100°F	
Aviation gasoline	6.1	5.7	6.0
JP-4	6.7	6.4	6.5
JP-5	7.2	6.8	6.8
JP-8	—	—	6.7

If fuel tanks of simple geometry are used, the tank volume can be calculated directly. Wing-box fuel volume can be approximated by assuming a tapered box shape. For complex integral and bladder tanks, the tank volume is determined using a fuel-volume plot as shown in Fig. 10.23. This is constructed by measuring the cross-sectional area of the tanks at various fuselage locations, then plotting those cross-sectional areas on a volume plot similar to the aircraft volume plot previously discussed.

If a discrete tank is used, the actually available internal volume can be calculated by subtracting the wall thickness from the external dimensions. For integral and bladder tanks, the available tank volume must be reduced from the measured value to allow for wall thickness, internal structure, and bladder thickness.

A rule of thumb is to assume that 85% of the volume measured to the external skin surface is usable for integral wing tanks, and 92% is usable for integral fuselage tanks. If bladder tanks are used, the values become 77% for wing tanks and 83% for fuselage tanks.

Note in Fig. 10.23 that the fuel volume plot allows the estimation of the center of gravity (c.g.) for each fuel tank, which is the centroid of the area plotted for the tank. The total fuel c.g. is simply the weighted average of the individual tank c.g.'s, and should be close to the aircraft c.g.

LANDING GEAR AND SUBSYSTEMS

11.1 INTRODUCTION

Of the many internal components that must be defined in an aircraft layout, the landing gear will usually cause the most trouble. Landing gear must be placed in the correct down position for landing, and must somehow retract into the aircraft without chopping up the structure, obliterating the fuel tanks, or bulging out into the slipstream. This chapter covers landing-gear design as well as installation of other subsystems.

11.2 LANDING GEAR ARRANGEMENTS

The common options for landing-gear arrangement are shown in Fig. 11.1. The single main gear is used for many sailplanes because of its simplicity. The wheel can be forward of the center of gravity (c.g.), as shown here, or can be aft of the c.g. with a skid under the cockpit.

"Bicycle" gear has two main wheels, fore and aft of the c.g., with small "outrigger" wheels on the wings to prevent the aircraft from tipping sideways. The bicycle landing gear has the aft wheel so far behind the c.g. that the aircraft must takeoff and land in a flat attitude, which limits this type of gear to aircraft with high lift at low angles of attack (i.e., high-aspect-ratio wings with large camber and/or flaps). Bicycle gear has been used mainly on aircraft with narrow fuselage and wide wing span such as the B-47.

The "taildragger" landing gear has two main wheels forward of the c.g. and an auxiliary wheel at the tail. Taildragger gear is also called "conventional" landing gear, because it was the most widely used arrangement during the first 40 years of aviation. Taildragger gear provides more propeller clearance, has less drag and weight, and allows the wing to generate more lift for rough-field operation than does tricycle gear.

However, taildragger landing gear is inherently unstable. If the aircraft starts to turn, the location of the c.g. behind the main gear causes the turn to get tighter until a "ground loop" is encountered, and the aircraft either drags a wingtip, collapses the landing gear, or runs off the side of the runway. To prevent this the pilot of a taildragger aircraft must align the aircraft almost perfectly with the runway at touchdown, and "dance" on the rudder pedals until the aircraft stops.

The most commonly used arrangement today is the "tricycle" gear, with two main wheels aft of the c.g. and an auxiliary wheel forward of the c.g.. With a tricycle landing gear, the c.g. is ahead of the main wheels so the aircraft is stable on the ground and can be landed at a fairly large "crab"